Unit BLDC Driver

SKU:U181













Description

Unit BLDC Driver is a driver unit specially designed for brushless DC (BLDC) motors. It adopts an STM32 MCU as the main controller and communicates via I2C to precisely control the DRV11873PWPR motor-driver chip, enabling PWM speed regulation, direction switching, and more. The driver chip features a simplified control interface and integrated stall protection, providing efficient and low-noise operation for small three-phase BLDC motors. The on-board STM32 upgrade interface facilitates firmware updates and functional expansion, making it ideal for industrial automation, smart robotics, precision instruments, and other fields.

Features

- STM32 MCU
- I2C communication (0x65)
- DRV11873 driver control
- BLDC (brushless DC motor)
- PWM speed configuration, direction switching, etc.
- Stall protection
- Firmware upgradable

Includes

- 1 x Unit-BLDC Driver
- 1 x HY2.0-4P Grove Cable (20cm)
- 1 x HT3.96-4P Terminal

Applications

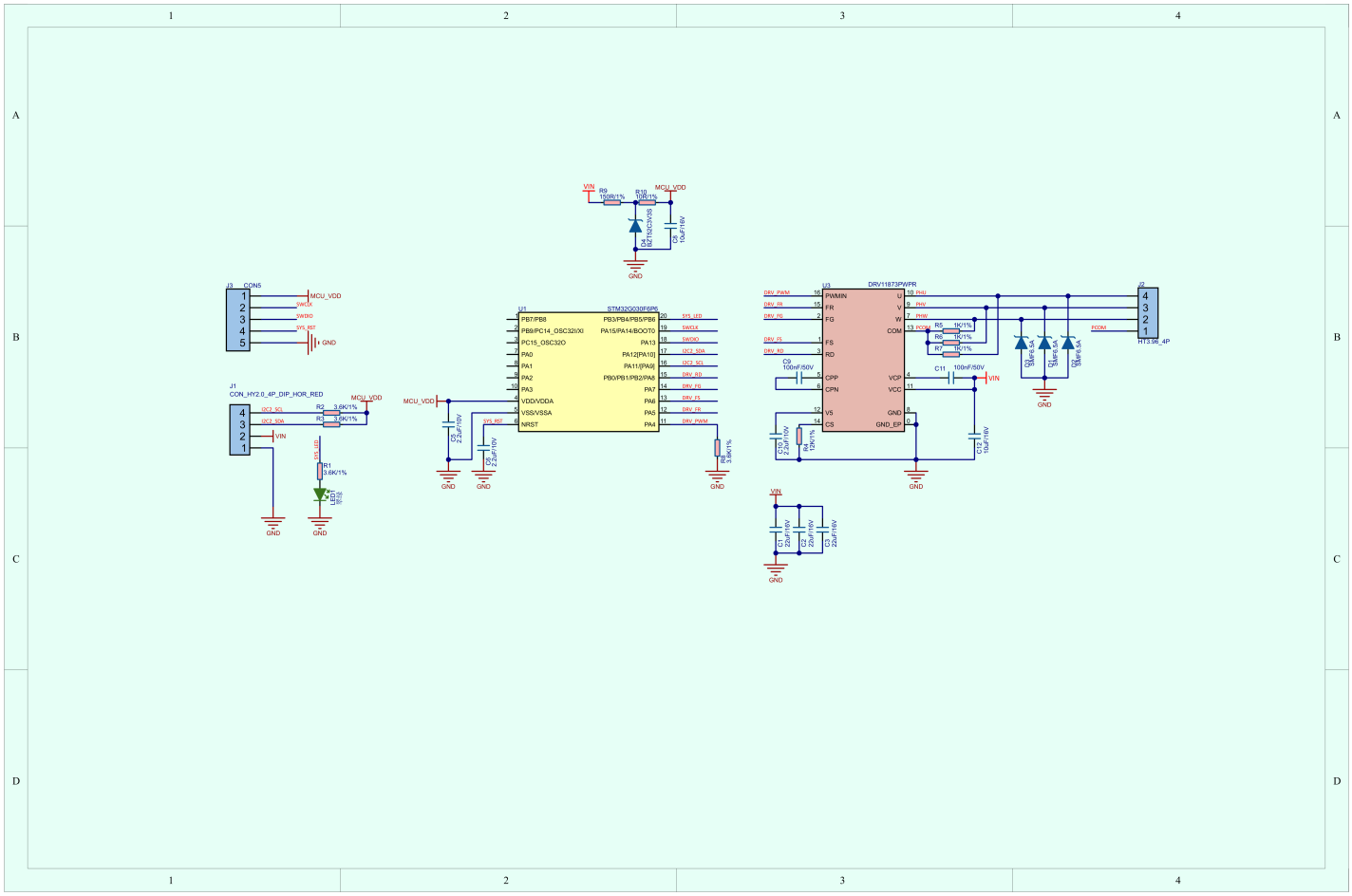
- Industrial Automation
- Smart Robotics
- Precision Instruments

Specifications

Specification	Parameter
MCU	STM32G030F6P6
Motor Driver Chip	DRV11873PWPR
Motor Type	Three-phase sensorless brushless DC (BLDC) motor
Motor Drive Voltage	5V (powered via Grove port)
Communication Port	I2C Communication @0x65
Output Load Capacity	DC 5V@508mA
Power Consumption	Standby Current: DC 5V@10.62mA Operating Current: DC 5V@13.20mA
Operating Temperature	0 ~ 40°C
Product Size	32.0 x 24.0 x 11.3mm
Product Weight	6.2g
Package Size	138.0 x 93.0 x 12.3mm
Gross Weight	13.4g

Schematics

- Unit BLDC Driver Schematics PDF



PinMap

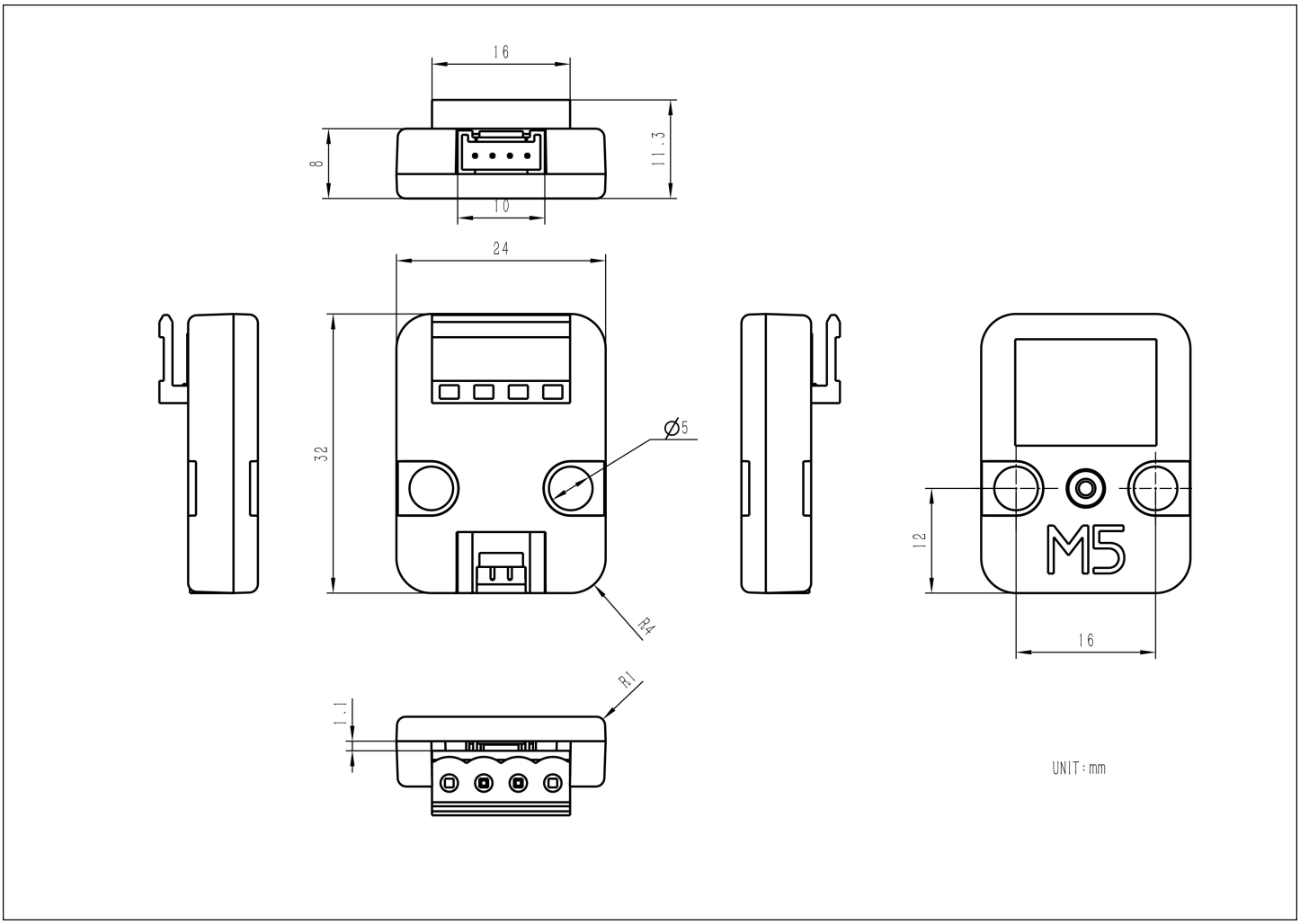
Unit BLDC Driver

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Indicator LED

STM32	PB3/PB4/PB5/PB6
Indicator LED (Green)	SYS_LED

Model Size



Structure

- Unit BLDC Driver Structure Files

Datasheets

- DRV11873

Softwares

Arduino

- Unit BLDC Driver I2C Addr Modify Example
- Unit-BLDC-Driver PWM Example
- Unit-BLDC-Driver RPM Example

UiFlow1

- Unit BLDC Driver UiFlow1 Docs

UiFlow2

- Unit BLDCDriver UiFlow2 Docs

Protocol

- Unit BLDC Driver I2C Protocol PDF

M5Stack Unit BLDC Driver I2C Protocol																		V1 (FW Version) 2023/12/5
REG MAP (Addr:0x65)			0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Setup	Mode	0x00 W/R	Mode																0: open loop; 1: closed loop (After switching modes, pwm or rpm will be automatically set to 0)
	Direction[1]	0x60 W/R	Direction																0~1 (You need to set PWM to 0 or RPM to 0 and then restart to take effect)
	Motor Model (Can be save to flash)	0x70 W/R	Motor model	Pole Pairs															Motor model: 0: Low Speed; 1: High Speed Pole Pairs: 1~255
	Motor Status	0x80 R	Motor Status																0: Standby; 1: Running; 2: Error
Open loop control	PWM	0x10 W/R	PWM-L	PWM-H															0~2047
Closed loop control	Readback RPM Float	0x20 R	Readback RPM-byte0	Readback RPM-byte1	Readback RPM-byte2	Readback RPM-byte3													float, get Motor Readback RPM
	Readback RPM X100 Int	0x90 R	Readback RPM-byte0	Readback RPM-byte1	Readback RPM-byte2	Readback RPM-byte3													Readback RPM X100 = Readback RPM-byte0 + Readback RPM-byte1 * 256 + Readback RPM-byte2 * 65536 + Readback RPM-byte3 * 16777216
	Readback RPM String	0xB0 R																	The string ends with '/0' and the maximum number of characters is 15
	Readback Pulse Frequency Float (Hz)	0x30 R	Readback Pulse Frequency -byte0	Readback Pulse Frequency -byte1	Readback Pulse Frequency -byte2	Readback Pulse Frequency -byte3													float, get Motor Readback Pulse Frequency RPM = Readback Pulse Frequency * 60 / Pole Pairs
	Readback Pulse Frequency X100 Int (Hz)	0xA0 R	Readback Pulse Frequency -byte0	Readback Pulse Frequency -byte1	Readback Pulse Frequency -byte2	Readback Pulse Frequency -byte3													Readback Pulse X100 = Readback Pulse-byte0 + Readback Pulse-byte1 * 256 + Readback Pulse-byte2 * 65536 + Readback Pulse-byte3 * 16777216
	Readback Pulse Frequency String (Hz)	0xC0 R																	The string ends with '/0' and the maximum number of characters is 15
	Setting RPM Float	0x40 W/R	Setting RPM-byte0	Setting RPM-byte1	Setting RPM-byte2	Setting RPM-byte3													float, setting target rpm (Only valid in closed loop)
	Setting RPM X100 Int	0xD0 W/R	Setting RPM-byte0	Setting RPM-byte1	Setting RPM-byte2	Setting RPM-byte3													Setting RPM X100 = Setting RPM-byte0 + Setting RPM-byte1 * 256 + Setting RPM-byte2 * 65536 + Setting RPM-byte3 * 16777216 (Only valid in closed loop)
	PID X100 (Can be save to flash)	0x50 W/R	P-byte0	P-byte1	P-byte2	P-byte3	I-byte0	I-byte1	I-byte2	I-byte3	D-byte0	D-byte1	D-byte2	D-byte3					P/I/D: PID = PID-byte0 + PID-byte1 * 256 + PID-byte2 * 65536 + PID-byte3 * 16777216 For example: P=50.05, P setting value=50.05*100=5005, P-byte0=0x8D, P-byte1=0x13, P-byte2=0, P-byte3=0
System	Flash Writeback	0xF0 W	Flash Writeback																Write 1 save to flash
	Firmware Version	0xF0 R															Version		Version: firmware version number
	I2C Address (Can be save to flash)	0xF0 R/W																Address	Address: 1~127

[1] Change direction:
Step1: Setup direction
Step2: Set PWM or RPM to 0
Step3: Set PWM or RPM to a value more than 0

Video

- Unit BLDC Driver product and function introduction

U181 BLDC Driver UNIT 视频.mp4

- UiFlow2 Unit BLDC Driver